

1. Which of the following functions are one-to-one (injective); which are onto (surjective)?

(a) $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x$

(b) $g : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ given by $g(x) = x^2$

(c) $h : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ given by $h(x) = 1 + \frac{1}{x}$

(d) $r : \mathbb{R} \rightarrow \mathbb{R}$ given by $r(x) = 3x^3 - 2$

2. Consider the function $f : \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$ given by $f(m, n) = (m + n, m + 2n)$.

(a) Show that f is an injective (that is, one-to-one) function.

(b) Show that f is a surjective (that is, onto) function.

3. A function $f : S \rightarrow T$ is called a *constant function* if there exists a $t \in T$ such that for all $s \in S$ $f(s) = t$.

(a) Is the following definition equivalent? Why or why not?

A function is called a *constant function* if for all $s \in S$ there exists a $t \in T$ such that $f(s) = t$.

Hint: It isn't.

For the rest of this problem, let $g : S \rightarrow T$ and $h : T \rightarrow S$ be functions, and assume their composition $h \circ g$ is a constant function.

- (b) Write out a well-quantified statement to explain what “assume their composition $h \circ g$ is a constant function” means.
- (c) What can you say about h given that g is onto? Express your claim as an if-then statement, and prove it.
- (d) What can you say about g given that h is one-to-one? Express your claim as an if-then statement, and prove it.
- (e) Is it necessarily true that $g \circ h$ is a constant function?

Functions – Answers / Solutions

1. (a) This is a bijection. We prove this in two steps, by showing first that f is injective and next that f is surjective.

To show that f is injective, we show that if $f(x_1) = f(x_2)$, then $x_1 = x_2$. But $f(x_1) = x_1$ and $f(x_2) = x_2$, so clearly these two statements are the same.

To show that f is surjective, we show that if $y \in \mathbb{R}$, then there is an $x \in \mathbb{R}$ with $f(x) = y$. This is simple: take $x = y$, so $f(y) = y$.

- (b) The function g is *not* injective, since $g(-a) = g(a) = a^2$. But g is surjective: If $y \in \mathbb{R}_{\geq 0}$ (that is, if $y \in \mathbb{R}$ and $y \geq 0$), then $y = g(\sqrt{y})$.

- (c) The function h is injective but not surjective.

To see injectivity, we assume that $h(x_1) = h(x_2)$ and prove that $x_1 = x_2$. If $h(x_1) = h(x_2)$, then $1 + \frac{1}{x_1} = 1 + \frac{1}{x_2}$, from which we find that $\frac{1}{x_1} = \frac{1}{x_2}$ and so $x_1 = x_2$.

We can *try* to prove surjectivity, and in failing we'll see why h is not surjective. Given $y \in \mathbb{R}$, we want to find x so that $h(x) = y$. Solving $1 + \frac{1}{x} = y$ for x , we get $x = \frac{1}{y-1}$. This isn't solvable when $y = 1$, and staring at $h(x) = 1 + \frac{1}{x}$ we can realize that $h(x)$ will never equal 1. Thus h is not surjective.

2. (a) Suppose $f(x, y) = f(a, b)$, then $(x + y, x + 2y) = (a + b, a + 2b)$. Thus we have the two equations

$$x + y = a + b$$

and

$$x + 2y = a + 2b$$

, and subtracting the first from the second we get $y = b$. Finally plugging into the first equation we get $x = a$. Thus $(a, b) = (x, y)$, so f is injective.

- (b) Let $(a, b) \in \mathbb{Z}^2$. We need to find (x, y) such that $f(x, y) = (a, b)$. Take

$$(x, y) = (2a - b, b - a),$$

then

$$f(x, y) = f(a - b, 2b - a) = (2a - b + b - a, 2a - b + 2(b - a)) = (a, b).$$

Thus f is surjective.

- (c) Suppose instead you consider

$$f : \mathbb{N}^2 \rightarrow \mathbb{N}^2$$

given by $f(x, y) = (x + y, x + 2y)$. Note f is no longer surjective since for all $(x, y) \in \mathbb{N}^2$, $f(x, y) \neq (1, 1)$.

3. (a) Here we are not making a single choice of t such that $f(x) = t$ for all $x \in S$; we are instead allowing different t for each element $x \in S$. This is just saying that $f(x) \in T$ for all $x \in S$, which we already knew from the definition of a function.

- (b) There exists an element $b \in S$ such that for all $x \in S$ we have $h(g(x)) = b$.

- (c) **Claim:** If g is surjective, then h is constant.

Proof. Since $h \circ g$ is a constant function, there exists an element $b \in S$ so that $(h \circ g)(x) = b$ for all $x \in S$. We want to prove that h is constant; namely we want to show that $h(y) = b$ for all $y \in T$.

Let $y \in T$. Since g is surjective we know there exists $a \in S$ so that $g(a) = y$. Now

$$h(y) = h(g(a)) = b$$

since $h \circ g$ is constant. Thus h must be a constant function when g is surjective. $\square$

(d) **Claim:** If h is injective, then g is constant.

Proof. Let $x, y \in S$, then we want to prove that $g(x) = g(y)$. Since $h \circ g$ is a constant function, we know that $h(g(x)) = h(g(y))$. Since h is injective, we have $g(x) = g(y)$. Thus g is a constant function. $\square$

(e) No, this is not true. Consider $S = \{a, b, c\} = T$ and $g(a) = a, g(b) = b, g(c) = b$ and $h(a) = a, h(b) = a, h(c) = b$, then $h \circ g$ is constant but $g \circ h$ is not.